COA 620 Bayesian Statistics in Ecology

Lecture 10 – Bayesian Multi-level Regression

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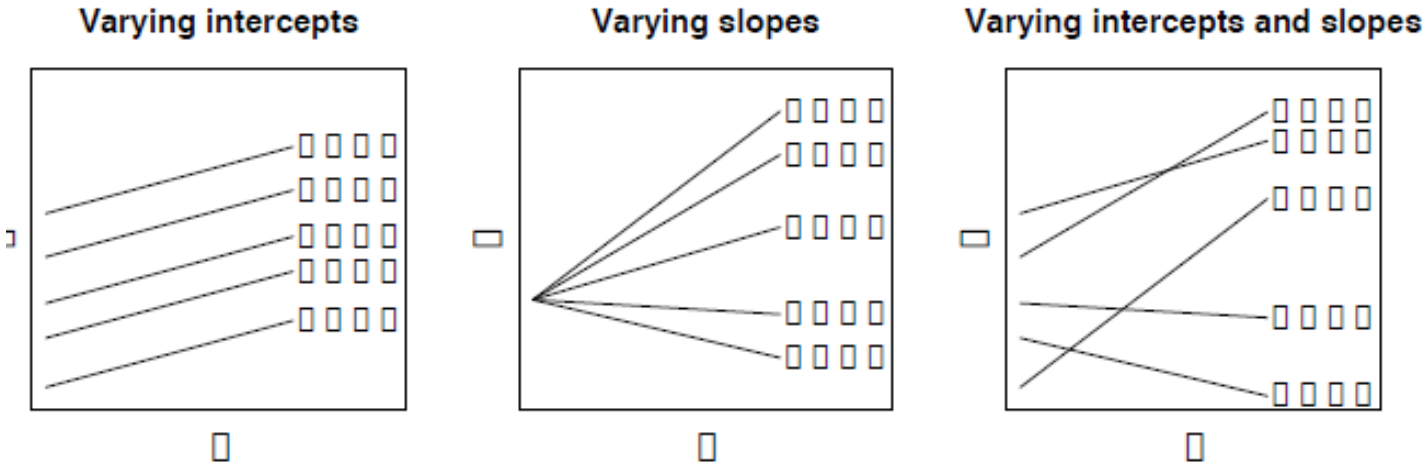
Multilevel models

Multi-level models are extensions of regression in which data are structured in groups and coefficients can vary by groups.

group level intercepts
group level intercepts with group level covariate
group level slopes and intercepts



The simple, Bayesian set-up

Deterministic model:

$$g(\boldsymbol{\theta}, x_i)$$

Stochastic model:

$$\underbrace{[\boldsymbol{\theta}, \sigma^2 | y_i]}_{\text{posterior}} \propto \overbrace{\underbrace{[y_i | g(\boldsymbol{\theta}, x_i), \sigma^2]}_{\text{likelihood}} \underbrace{[\boldsymbol{\theta}]}_{\text{priors}}}^{\text{joint}}$$

Draw the DAG.

Recall that

$$\underbrace{[\boldsymbol{\theta}, \sigma^2 | y_i]}_{\text{posterior}} \propto \overbrace{[y_i, \boldsymbol{\theta}, \sigma^2]}^{\text{joint}}$$

Hierarchical models: “ modeling parameters”

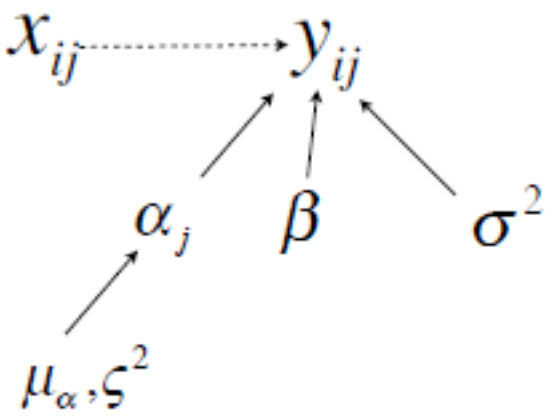
$$
\begin{aligned}
[\theta_1, \boldsymbol{\theta}_2, \boldsymbol{\alpha}, y_i, \sigma^2] \quad &\propto \quad [y_{ij} | g(\theta_1, \theta_{2,j}, x_{ij}), \sigma_1^2] \\
&\times \quad [\theta_{2,j} | h(\alpha_1, \alpha_2, u_j), \sigma_2^2] \\
&\times \quad [\theta_1, \boldsymbol{\theta}_2, \boldsymbol{\alpha}, \boldsymbol{\sigma}^2]
\end{aligned}
$$

Draw the DAG.

The problem

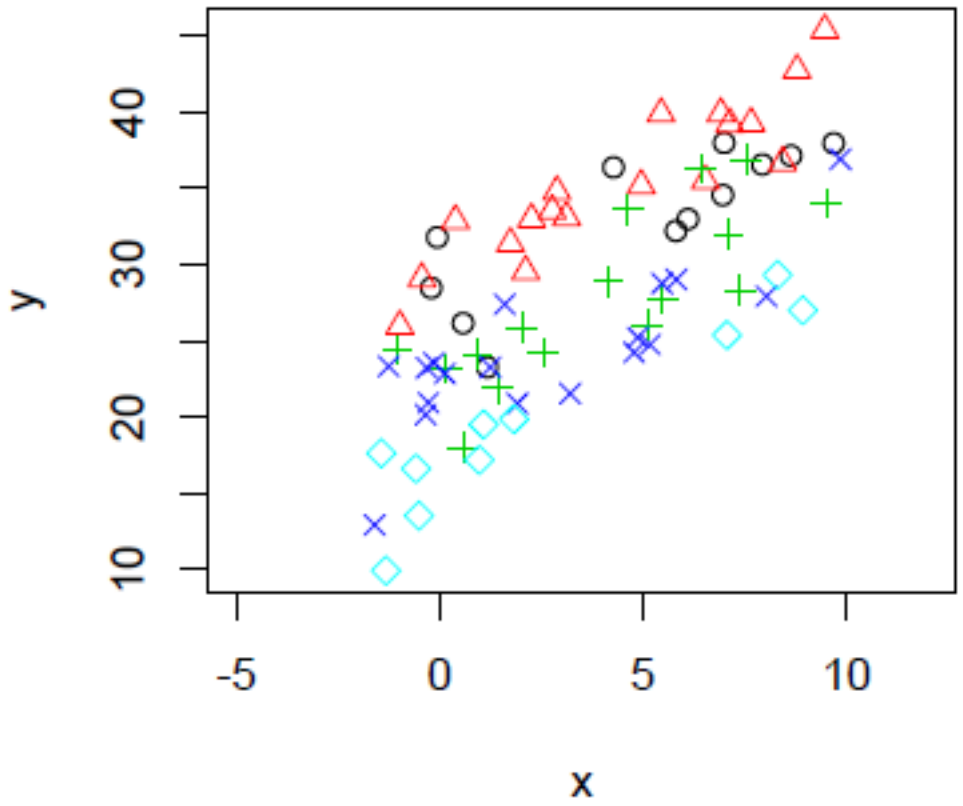
Group 1	Group 2	Group 3		Group n
$\mathbf{y}_1, \mathbf{x}_1$	$\mathbf{y}_2, \mathbf{x}_2$	$\mathbf{y}_3, \mathbf{x}_3$	.....	$\mathbf{y}_n, \mathbf{x}_n$

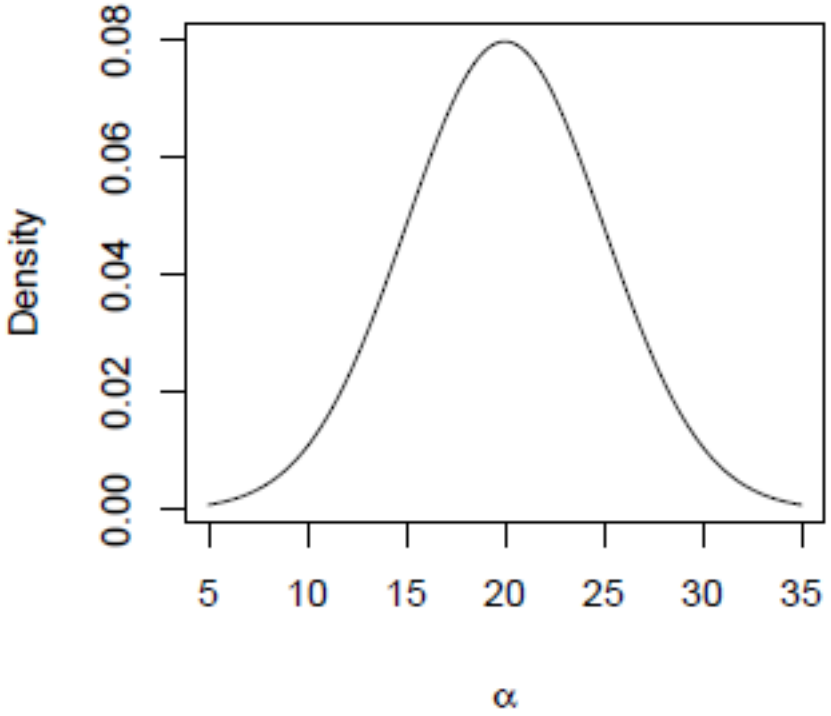
We can model the intercept:



$$
\begin{aligned}
&[\beta, \boldsymbol{\alpha}, \sigma^2, \mu_\alpha, \varsigma^2, |\mathbf{y}] \propto \prod_{i=1}^{n_j} \prod_{j=1}^{J} \text{normal}\left(y_{ij} | \alpha_j + \beta x_{ij}, \sigma^2\right) \\
&\times \text{normal}\left(\alpha_j | \mu_\alpha, \varsigma^2\right) \\
&\times \text{normal}\left(\beta | 0, 10000\right) \text{normal}\left(\mu_\alpha | 0, 1000\right) \\
&\times \text{inverse gamma}\left(\sigma^2 | .001, .001\right) \text{inverse gamma}\left(\varsigma^2 | .001, .001\right)
\end{aligned}
$$

We seek to understand the distribution of intercepts.





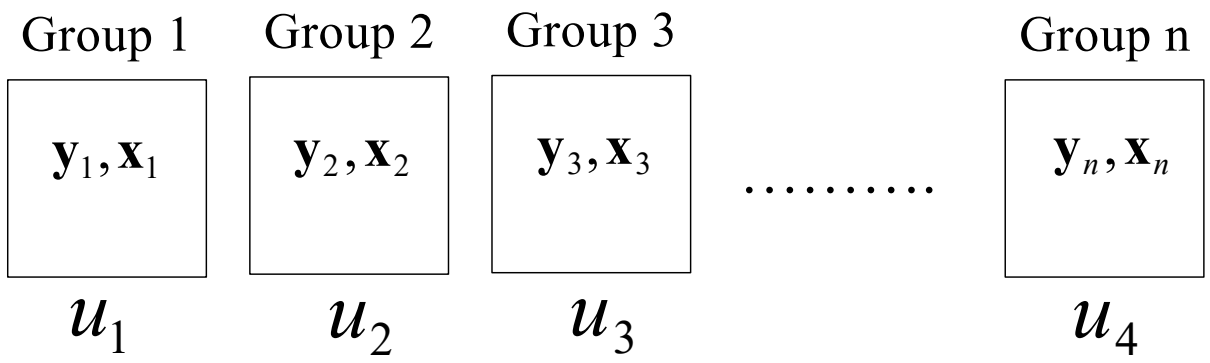
Some notation

$$\mu_{ij} = \beta_0 + \beta_1 x_{ij} + \alpha_j$$
$$y_{ij} \sim \text{normal}(\mu_{ij}, \sigma^2)$$
$$\alpha_j \sim \text{normal}(0, \varsigma^2)$$

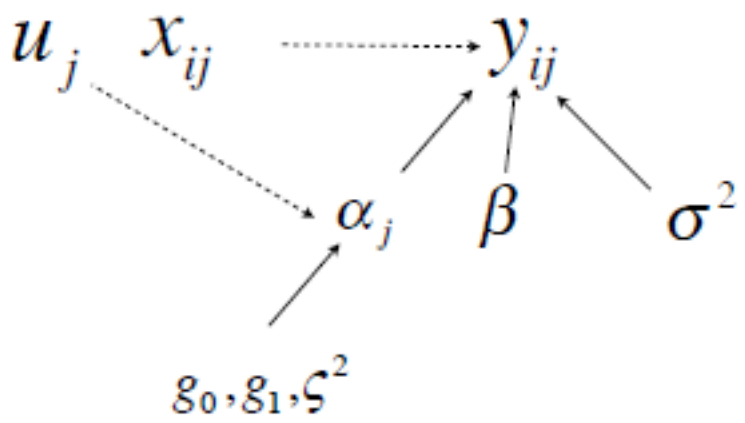
is identical to:

$$\mu_{ij} = \alpha_j + \beta_1 x_{ij}$$
$$y_{ij} \sim \text{normal}(\mu_{ij}, \sigma^2)$$
$$\alpha_j \sim (\mu_\alpha, \varsigma^2)$$

Include data on groups.



We can model the intercept as a function of group level data:

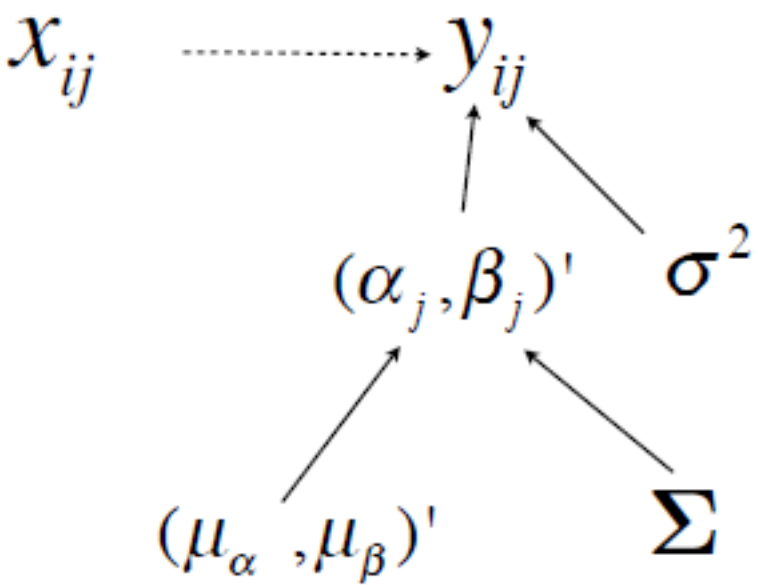


$$
\begin{aligned}
&\left[\boldsymbol{\alpha}, \beta, \sigma^2, \mathrm{g}, \varsigma^2, |\mathrm{y}\right] \propto \prod_{i=1}^{n_j} \prod_{j=1}^{J} \mathrm{normal}\left(y_{ij} | \alpha_j + \beta x_{ij}, \sigma^2\right) \\
&\times \mathrm{normal}\left(\alpha_j | g_0 + g_1 u_j, \varsigma^2\right) \\
&\times \mathrm{normal}\left(\beta | 0, .001\right) \mathrm{normal}\left(g_0 | 0, 1000\right) \mathrm{normal}\left(g_1 | 0, 1000\right) \\
&\times \mathrm{inverse\,gamma}\left(\sigma^2 | .001, .001\right) \mathrm{inverse\,gamma}\left(\varsigma^2 | .001, .001\right)
\end{aligned}
$$

Indexing groups

```
model{
beta ~ dnorm(0,.0001)
sigma ~ dunif(0,50)
tau.p <- 1/sigma^2
g0 ~ dnorm(0,.0001)
g1 ~ dnorm(0,.0001)
varsigma ~ dunif(0,50)
tau.g <- 1/varsigma^2
 for (i in 1:length(y)){
  mu[i] <- alpha[group[i]]+ beta*x[i]
  y[i] ~ dnorm(mu[i],tau.p)
  }
  for(j in 1:n.group){
  mu.g[j] <- g0 + g1*u[j]
  alpha[j]~dnorm(mu.g[j],tau.g)
    }
```

Modeling intercepts *and* slopes



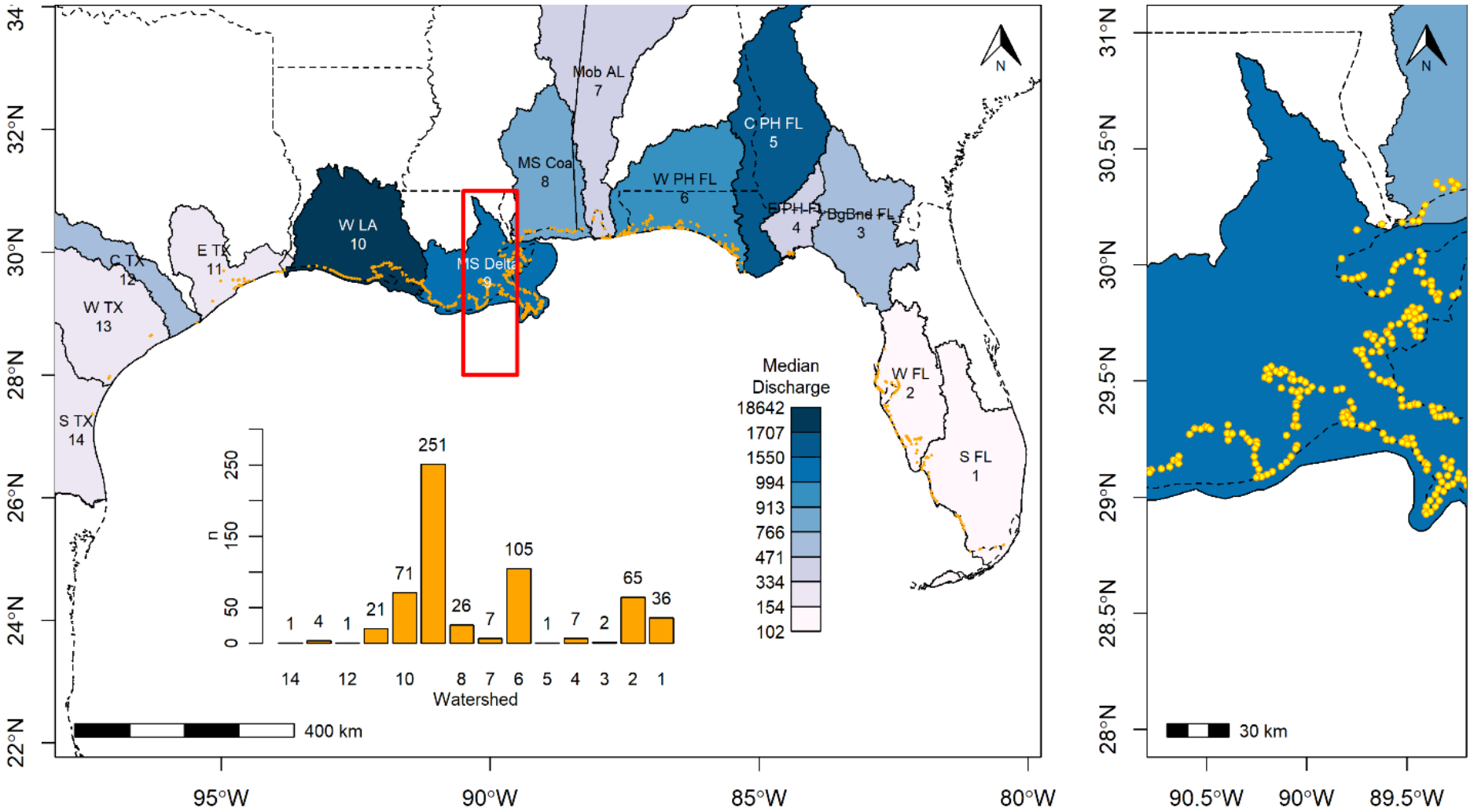
$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim \text{multivariate normal}\left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \boldsymbol{\Sigma} \right)$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} \varsigma_\alpha^2 & \text{Cov}(\boldsymbol{\alpha}, \boldsymbol{\beta}) \\ \text{Cov}(\boldsymbol{\alpha}, \boldsymbol{\beta}) & \varsigma_\beta^2 \end{pmatrix}, \text{Cov}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \rho \varsigma_\alpha \varsigma_\beta$$

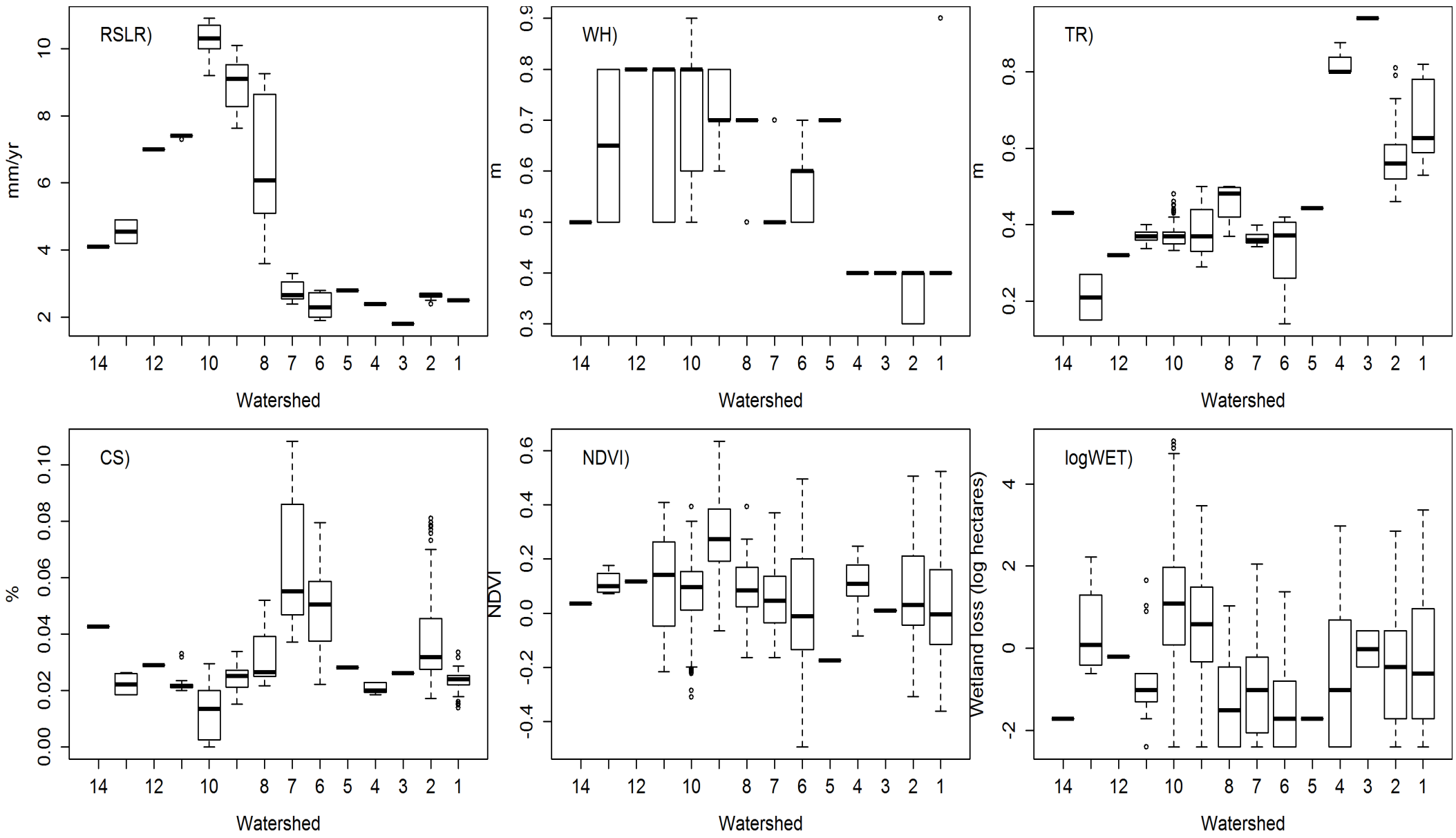
Modeling intercepts *and* slopes

$$
\begin{aligned}
[\boldsymbol{\alpha}, \boldsymbol{\beta}, \mu_\alpha, \mu_\beta, \sigma^2, \varsigma_\alpha, \varsigma_\beta, \rho | \mathbf{y}] \quad \propto \quad & \prod_{j=1}^{J} \prod_{i=1}^{n_j} \text{normal}(y_{ij} | \alpha_j + \beta_j x_{ij}, \sigma^2) \\
\times \quad & \text{MVN}\left(\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \middle| \begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \boldsymbol{\Sigma} \right) \\
\times \quad & \text{priors on } \mu_\alpha, \mu_\beta, \sigma^2, \varsigma_\alpha, \varsigma_\beta, \rho
\end{aligned}
$$

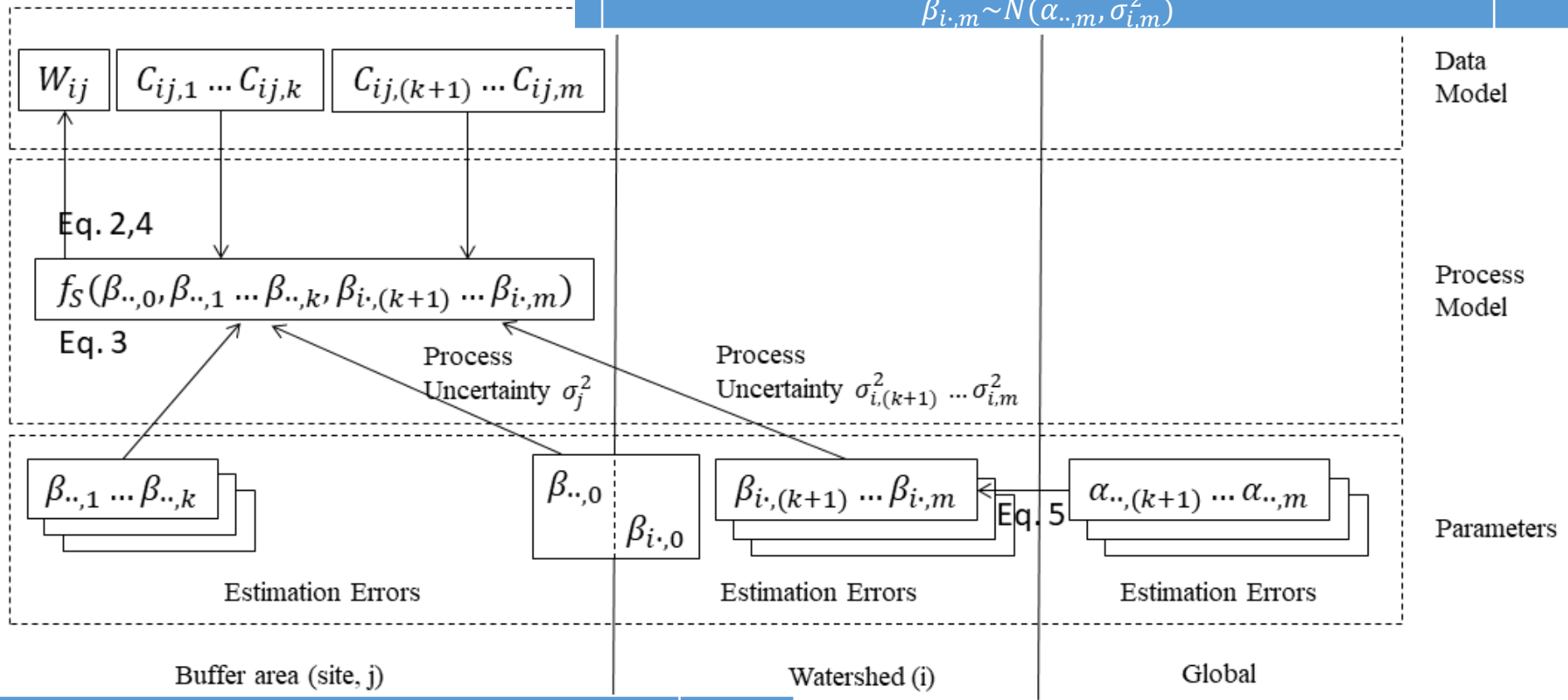
Modeling spatial variability of resilience of coastal wetlands to sea-level rise in the northern Gulf of Mexico using Bayesian multi-level models


34°N
32°N
30°N
28°N
26°N
24°N
22°N
95°W
90°W
85°W
80°W
N
Mob AL
7
MS Coa
8
C PH FL
5
W PH FL
6
EPH FL
4
BgBnd FL
3
W LA
10
MS Delta
9
E TX
11
C TX
12
W TX
13
S TX
14
W FL
2
S FL
1
Median
Discharge
18642
1707
1550
994
913
766
471
334
154
102
n
0
50
150
250
1
4
1
21
71
251
26
7
105
1
7
2
65
36
14
12
10
8
7
6
5
4
3
2
1
Watershed
400 km

31°N
30.5°N
30°N
29.5°N
29°N
28.5°N
28°N
90.5°W
90°W
89.5°W
N
30 km


RSLR)
mm/yr
WH)
m
TR)
m
CS)
%
NDVI)
NDVI
logWET)
Wetland loss (log hectares)
Watershed
14 12 10 8 7 6 5 4 3 2 1

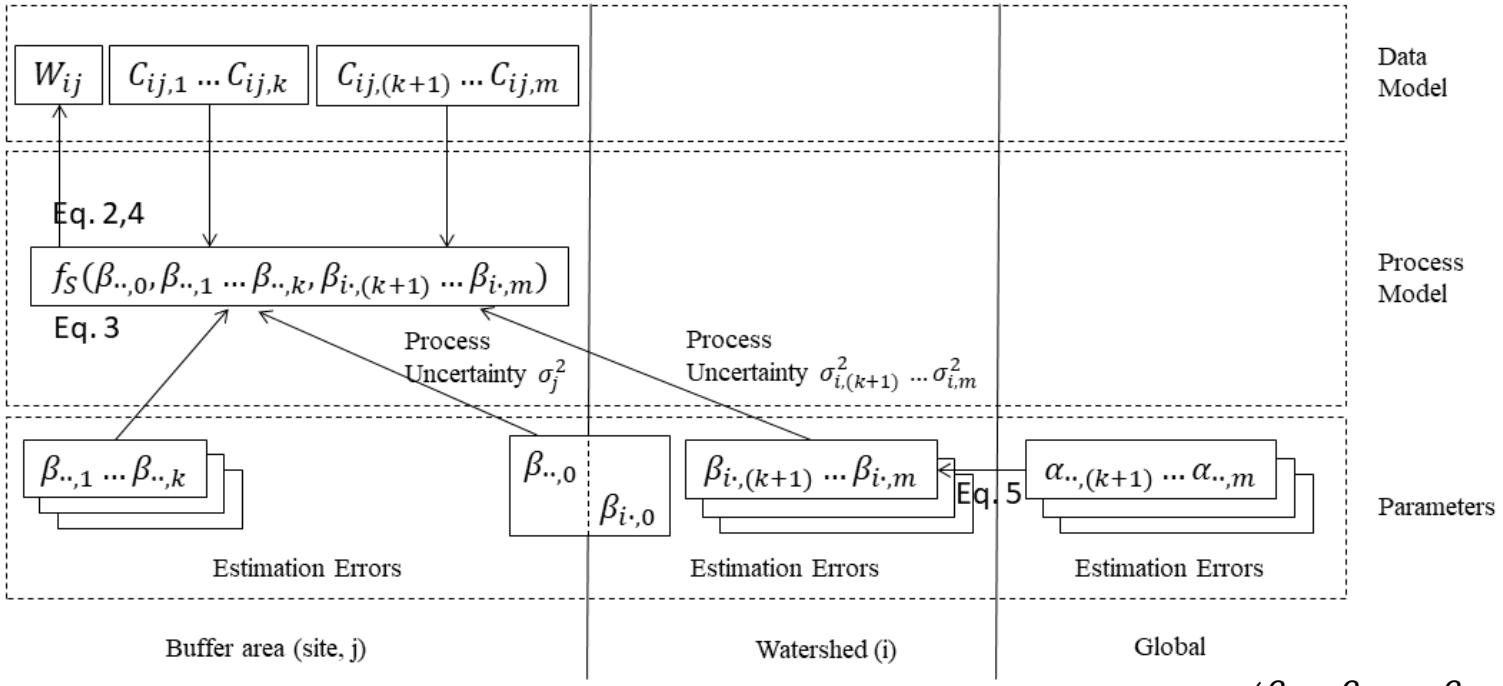
$$\beta_{i\cdot,(k+1)} \sim N(\alpha_{\cdot\cdot,(k+1)}, \sigma^2_{i,(k+1)})$$
$$\beta_{i\cdot,(k+2)} \sim N(\alpha_{\cdot\cdot,(k+2)}, \sigma^2_{i,(k+2)})$$
$$\dots$$
$$\beta_{i\cdot,m} \sim N(\alpha_{\cdot\cdot,m}, \sigma^2_{i,m}) \quad (5)$$



$$W_{ij} \sim N(W_{ij}.\mu, \sigma_j^2). \quad (2)$$

$$W_{ij}.\mu = f(\beta_{\cdot\cdot,0}, \beta_{\cdot\cdot,1} \dots \beta_{\cdot\cdot,k}, \beta_{i\cdot,(k+1)} \dots \beta_{i\cdot,m})$$
$$= \beta_{\cdot\cdot,0} + \sum_{l=1}^{k} \beta_{\cdot\cdot,l} C_{ij,l} + \sum_{l=k+1}^{m} \beta_{i\cdot,l} C_{ij,l} \quad (3)$$

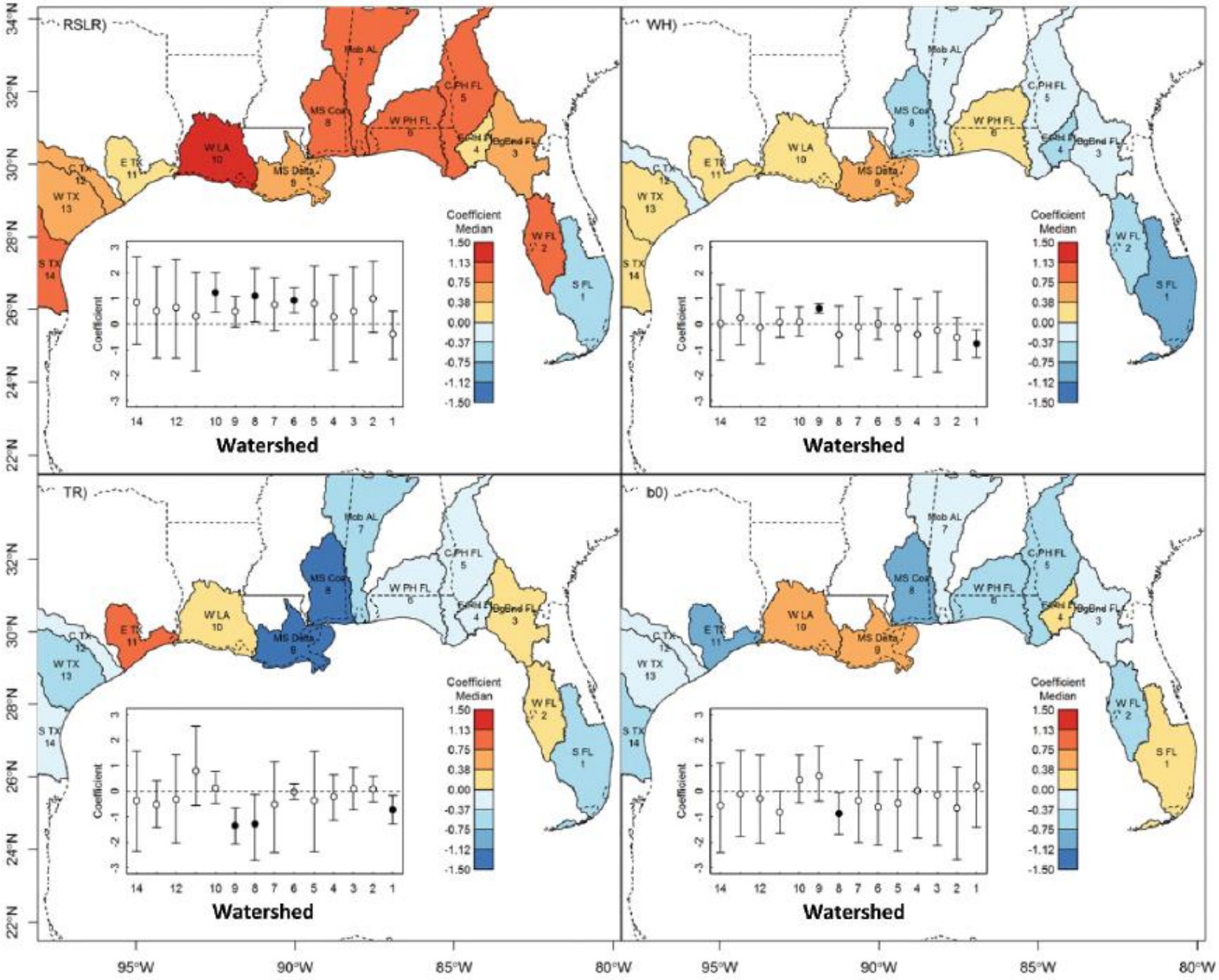
$$p(W | \beta_{\cdot\cdot,0}, \beta_{\cdot\cdot,1} \dots \beta_{\cdot\cdot,k}, \beta_{i\cdot,(k+1)} \dots \beta_{i\cdot,m})$$
$$\propto \prod_{i=1}^{R} \prod_{j=1}^{S_i} N(W_{ij} | f(\beta_{\cdot\cdot,0}, \beta_{\cdot\cdot,1} \dots \beta_{\cdot\cdot,k}, \beta_{i\cdot,(k+1)} \dots \beta_{i\cdot,m}), \sigma_j^2) \quad (4)$$



$$p\left(\begin{matrix}\beta_{\cdot\cdot,0}, \beta_{\cdot\cdot,1} \dots \beta_{\cdot\cdot,k}, \beta_{i\cdot,(k+1)} \dots \beta_{i\cdot,m} \\ \alpha_{\cdot\cdot,(k+1)} \dots \alpha_{\cdot\cdot,m}, \\ \sigma_j^2, \sigma_{i,(k+1)}^2, \dots, \sigma_{i,m}^2\end{matrix} \middle| W, C_1 \dots C_m\right)$$

$$\propto p\left(WL \middle| \beta_{\cdot\cdot,0}, \beta_{\cdot\cdot,1}, \dots, \beta_{\cdot\cdot,k}, \beta_{i\cdot,(k+1)}, \dots, \beta_{i\cdot,m}, \sigma_j^2\right)$$
$$\times \left(p\left(\beta_{i\cdot,(k+1)} \middle| \alpha_{\cdot\cdot,(k+1)}, \sigma_{i,(k+1)}^2\right) \times \dots \times p\left(\beta_{i\cdot,m} \middle| \alpha_{\cdot\cdot,m}, \sigma_{i,m}^2\right)\right) \tag{6}$$

$$\left.\begin{aligned}&\times\ p(\beta_{\cdot\cdot,0}) \times \left(p(\beta_{\cdot\cdot,1}) \times \dots \times p(\beta_{\cdot\cdot,k})\right) \times p(\sigma_j^2) \\ &\times \left(p(\alpha_{\cdot\cdot,(k+1)}) \times \dots \times p(\alpha_{\cdot\cdot,m})\right) \\ &\times \left(p(\sigma_{i,(k+1)}^2) \times \dots \times p(\sigma_{i,m}^2)\right)\end{aligned}\right\} \text{Priors}$$


RSLR)
WH)
TR)
b0)
Coefficient Median
1.50
1.13
0.75
0.38
0.00
-0.37
-0.75
-1.12
-1.50
Coefficient
Watershed
14 12 10 9 8 7 6 5 4 3 2 1
Mob AL 7
MS Coa 8
W PH FL 6
C PH FL 5
E PH FL 4
BgBnd FL 3
W FL 2
S FL 1
W LA 10
MS Delta 9
E TX 11
C TX 12
W TX 13
S TX 14
34°N
32°N
30°N
28°N
26°N
24°N
22°N
95°W
90°W
85°W
80°W